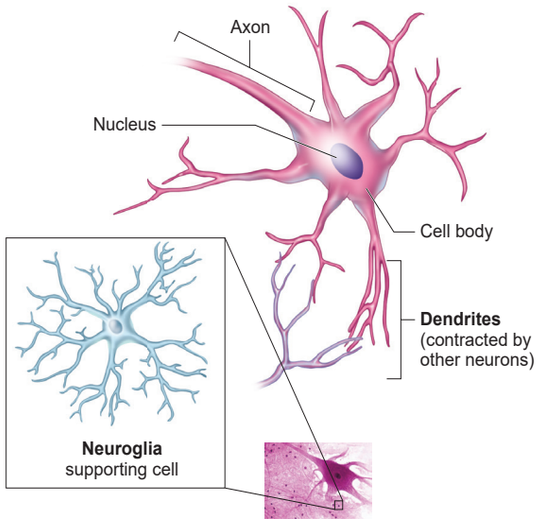


Figure 3.8 Nervous tissues with neuroglia

conditions. Neurons, the unit of neural system are excitable cells (Figure 3.8). The neuroglial cells which constitute the rest of the neural system protect and support the neurons. **Neuroglia** makes up more than one-half of the volume of neural tissue in our body.

When a neuron is suitably stimulated, an electrical disturbance is generated which swiftly travels along its plasma membrane. Arrival of the disturbance at the neuron's endings, or output zone, triggers events that may cause stimulation or inhibition of adjacent neurons and other cells (You will study in detail in Chapter 10)

Diseases of Nervous System:

1. **Parkinson's disease:** A degenerative disorder of the nervous system that affects movement, often including tremors.
2. **Alzheimer's disease:** It is a chronic neurodegenerative disease which includes the symptoms of difficulty in remembering recent events, problems with language, disorientation and mood swings.

Biopsy is an examination of tissue or liquid removed from a living body to discover the presence, cause or extent of a disease.

Autopsy is a post-mortem (dissection of a dead body) examination to discover the cause of death or the extent of disease.

The field of **Forensic science** effectively uses the histological techniques to trace out crimes.

Summary

The body cells combine to form four different types of tissues; epithelial, connective, muscle and nervous tissues. Though the cells of these tissues share certain features in common, by no means they are identical. They belong together because they have basic fundamental resemblances. The important concept to carry away with you is that tissues, despite their unique abilities, cooperate to keep the body safe, healthy, viable and whole.

Activity

1.Students are asked to identify the unlabelled slides of tissues and to classify them. Similar exercise can also be accomplished by projecting unlabelled histological images on a screen. They can identify the slides of different tissues through microscope

2.The preparation of smear of stratified squamous epithelia from the inner lining of cheek allows the students to make their own slides using biological stain. They will have the experience of examining their cheek cells.

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